

# Analytical and Finite-Population Properties of Defector-Responsive Strategy Switching

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## Abstract

Defector-responsive switching from unconditional cooperation to Tit-for-Tat has been proposed as a rescue mechanism in the repeated Prisoner's Dilemma. The response rule itself was presented by Pal and coauthors; here we characterize analytical and finite-population consequences that were not established there. We derive an exact break-even readiness cost for general benefit and cooperation-cost parameters and state the existence and invasion conditions under which the comparison is valid. We then formulate a continuous-time finite-population process with separate birth-death and individual strategy-transition events. Its infinitesimal drift converges to the deterministic replicator-switching equations, resolving the timescale and interpretation problems created when switching is attached to reproduction. Simulations show that constant switching produces Tit-for-Tat rapidly but continues to remove unconditional cooperators after defectors disappear. Responsive switching can rescue slightly less reliably at low intensity, but it shuts off after rescue and preserves unconditional cooperation over 10 to 100 demographic generations. This preservation advantage remains positive across 27 combinations of population size, selection strength, and benefit-to-cost ratio. Adding the readiness cost to the stochastic process reduces the responsive advantage continuously but does not erase it over the tested finite horizons. The results provide a deterministic-stochastic correspondence, a quantitative cost constraint, and a robust finite-population tradeoff for an existing feedback mechanism.

**Keywords:** evolutionary game theory, repeated Prisoner's Dilemma, state-dependent switching, continuous-time Moran process, cooperation, feedback dynamics

# 1 Introduction

The Prisoner’s Dilemma expresses a conflict between individual and collective benefit: defection is individually favored in a single encounter, although mutual cooperation yields the larger joint return. Repeated interaction can change the evolutionary outcome because strategies may condition present behavior on earlier actions (Axelrod and Hamilton 1981; Axelrod 1984; Nowak 2006). Tit-for-Tat (TFT), which cooperates initially and then copies its opponent’s preceding action, can punish unconditional defection while continuing to cooperate with unconditional cooperators (Nowak and Sigmund 1992). The repeated game supports a much broader space of conditional and memory-dependent strategies (Stewart and Plotkin 2013).

Evolutionary game dynamics provide a natural framework for asking whether conditional behavior can protect a population dominated by unconditional cooperation. Replicator and selection-mutation equations describe the corresponding infinite-population dynamics (Taylor and Jonker 1978; Hofbauer and Sigmund 1998; Hofbauer 1985; Haderler 1981; Page and Nowak 2002). Mutation can change both the number and stability of equilibria, including in multi-strategy and noisy social dilemmas (Antal et al. 2009; Chen et al. 2024). Finite-population formulations add demographic stochasticity and distinguish invasion, extinction, and fixation events (Moran 1962; Nowak et al. 2004; Traulsen et al. 2006; Fudenberg and Imhof 2006). In repeated games, mutation can reorganize cycles of cooperation and defection or promote diverse cooperative communities (Imhof et al. 2005, 2007; Tkadlec et al. 2023).

Pal et al. (2025) studied a rescue mechanism with three strategies: unconditional cooperation (ALLC), TFT, and unconditional defection (ALLD). In their principal deterministic model, switching from ALLC to TFT occurs at a constant rate. TFT can suppress an ALLD invasion, but constant switching also continues after the invasion has ended. Pal et al. also considered switching proportional to ALLD abundance in their electronic supplementary material, Figure S8. The feedback rule used here is therefore not claimed as a new mechanism.

Our contribution is to characterize three consequences of that rule. First, we add a payoff cost for maintaining a defector-responsive switching capability. We derive the exact cost at which the cooperative boundary equilibrium matches its constant-switching counterpart for arbitrary benefit  $b$  and cooperation cost  $c$ , together with the stability restrictions required for the comparison. Second, we construct a continuous-time finite-population process whose individual switching events are separate from reproduction and whose large-population drift equals the deterministic equations. This provides a direct deterministic-stochastic correspondence rather than an update-dependent rate-to-probability conversion. Third, we test the resulting tradeoff across post-rescue horizons, population sizes, selection strengths, benefit-to-cost ratios, and readiness costs.

This state-dependent transition belongs to a broader class of feedback in evolutionary dynamics. State-dependent mutation can alter long-run selection (Bergin and Lipman 1996); game-environment feedback can make strategies and environmental states coevolve (Weitz et al. 2016); and perception of social information can enter evolutionary decision rules directly (Salahshour 2025). The present model is narrower: the

current ALLD frequency changes the transition intensity between two fixed strategies, not the payoff matrix or the perceived action itself.

## 2 Model

### 2.1 Repeated-game payoffs

Let  $x$ ,  $y$ , and  $z$  be the frequencies of ALLC, TFT, and ALLD, respectively, with  $x + y + z = 1$ . We use the payoff convention of [Pal et al. \(2025\)](#):

$$\begin{array}{c|ccc}
 & \text{ALLC} & \text{TFT} & \text{ALLD} \\
 \hline
 \text{ALLC} & b - c & b - c & -c \\
 \text{TFT} & b - c & (b - c)/2 & 0 \\
 \text{ALLD} & b & 0 & 0
 \end{array} \quad (1)$$

This matrix comes from an infinitely repeated donation game in the limit of rare implementation error. An intended cooperation is implemented as defection with probability  $\alpha\epsilon$ , and an intended defection is implemented as cooperation with probability  $\beta\epsilon$ . Payoffs are long-run averages under the stationary distribution of the four outcomes  $CC$ ,  $CD$ ,  $DC$ , and  $DD$ , followed by the limit  $\epsilon \rightarrow 0^+$ . Pal et al. take symmetric errors,  $\alpha = \beta$ . In a TFT-TFT match the four outcomes then have equal limiting stationary weight, giving

$$\frac{(b - c) + (-c) + b + 0}{4} = \frac{b - c}{2}. \quad (2)$$

Thus, the TFT-TFT entry is not the noiseless value. Without implementation error, mutual TFT beginning with cooperation would instead receive  $b - c$  per round.

Writing  $q = b - c > 0$ , the expected payoffs are

$$f_C = q(x + y) - cz, \quad f_T = q(x + y/2), \quad f_D = bx, \quad (3)$$

and  $\bar{f} = xf_C + yf_T + zf_D$ .

### 2.2 Deterministic switching dynamics

One-way switching from ALLC to TFT gives

$$\begin{aligned}
 \dot{x} &= x(f_C - \bar{f}) - \mu x, \\
 \dot{y} &= y(f_T - \bar{f}) + \mu x, \\
 \dot{z} &= z(f_D - \bar{f}).
 \end{aligned} \quad (4)$$

We compare a constant rule,  $\mu = \mu_0$ , with the defector-responsive rule,  $\mu = \mu_0 z$ .

To represent the cost of maintaining the responsive capability, the payoff of ALLC is reduced by  $k \geq 0$ . Thus,  $f_C$  is replaced by  $f_C - k$  in its replicator term and the population mean becomes  $\bar{f}_k = \bar{f} - kx$ . The switching flow remains  $\mu_0 zx$ . The cost is state-specific: it represents maintaining the latent ALLC program in readiness to detect defectors and enter the TFT phenotype. After an individual switches, it has left

that latent state and no longer pays this particular readiness cost. A persistent sensing cost paid by both ALLC and TFT would be a different model and would remove the interior ALLC-TFT boundary balance derived below.

### 2.3 Continuous-time finite-population process

The finite model is a continuous-time Moran-type birth-death process with a separate phenotypic transition channel. Let  $\mathbf{n} = (n_C, n_T, n_D)$ ,  $N = n_C + n_T + n_D$ , and  $\mathbf{X}^N = \mathbf{n}/N$ . Expected payoffs  $\pi_i^N$  are calculated against the other  $N - 1$  individuals using (1). In the responsive costly model,  $k$  is subtracted from  $\pi_C^N$  only. An individual of type  $i$  gives birth at rate

$$r_i = a + w\pi_i^N, \quad a = 3, \quad (5)$$

and its offspring replaces a uniformly selected individual. The common baseline  $a$  keeps all birth rates nonnegative over the reported parameter ranges and cancels from the deterministic selection drift. The parameter  $w$  controls selection strength.

Independently of reproduction, each ALLC individual changes phenotype to TFT at intensity  $\mu_0$  under constant switching and  $\mu_0 n_D/N$  under responsive switching. The corresponding population-level transition rates are

$$q_{C \rightarrow T}^N = \mu_0 n_C \quad \text{and} \quad q_{C \rightarrow T}^N = \mu_0 \frac{n_D}{N} n_C, \quad (6)$$

respectively. A transition changes an existing individual, rather than the type of a newborn. Consequently, it represents behavioral or phenotypic switching, not mutation or transmission error.

**Proposition 1** (Deterministic limit) *Let  $h(z) = 1$  for constant switching and  $h(z) = z$  for responsive switching. For a smooth test function  $\varphi$ , the generator of the frequency process is*

$$\begin{aligned} \mathcal{L}_N \varphi(\mathbf{x}) = & N \sum_{i,j} x_i x_j r_i \left[ \varphi\left(\mathbf{x} + \frac{\mathbf{e}_i - \mathbf{e}_j}{N}\right) - \varphi(\mathbf{x}) \right] \\ & + N \mu_0 x h(z) \left[ \varphi\left(\mathbf{x} + \frac{\mathbf{e}_T - \mathbf{e}_C}{N}\right) - \varphi(\mathbf{x}) \right]. \end{aligned} \quad (7)$$

*Its infinitesimal drift converges as  $N \rightarrow \infty$  to*

$$w \begin{pmatrix} x(\tilde{f}_C - \bar{f}) \\ y(f_T - \bar{f}) \\ z(f_D - \bar{f}) \end{pmatrix} + \mu_0 x h(z) \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad (8)$$

*where  $\tilde{f}_C = f_C - k$  in the costly responsive model,  $\tilde{f}_C = f_C$  otherwise, and  $\bar{f} = x\tilde{f}_C + yf_T + zf_D$ . For  $w = 1$ , (8) is exactly (4), including its switching flux.*

*Proof.* For coordinate  $i$ , a birth-death event contributes

$$\sum_j x_i x_j (r_i - r_j) = x_i (r_i - \bar{r}) = w x_i (\tilde{f}_i - \bar{f}), \quad (9)$$

because the common baseline  $a$  cancels. The transition in (6) contributes  $\mu_0 x h(z)(-1, 1, 0)^\top$ . Finite-population payoffs calculated without self-interaction converge uniformly to (3). Standard fluid-limit results for density-dependent jump processes then give convergence on finite time intervals under the usual regularity conditions (Kurtz 1970).  $\square$

This construction fixes the time unit explicitly: payoffs determine individual birth rates through (5), while  $\mu_0$  is an individual transition intensity in the same continuous-time clock. No unit-length interval is inserted into a probability transformation.

### 3 Analytical Cost Result

The boundary  $z = 0$  permits a direct comparison of the two switching rules. It is also where their difference is clearest: constant switching remains active, whereas defector-responsive switching stops.

**Proposition 2** (Break-even readiness cost) *Assume  $b > c > 0$  and write  $q = b - c$ .*

1. *Under constant switching, the nontrivial ALLC-TFT boundary equilibrium is*

$$x_F^* = 1 - \sqrt{\frac{2\mu_0}{q}}, \quad y_F^* = \sqrt{\frac{2\mu_0}{q}}, \quad (10)$$

*and exists for  $0 \leq \mu_0 \leq q/2$ . Its transverse ALLD invasion exponent is*

$$g_F(\mu_0) = c - b\sqrt{\frac{2\mu_0}{q}} + \mu_0. \quad (11)$$

2. *Under defector-responsive switching with readiness cost  $k$ , the nontrivial boundary equilibrium is*

$$x_R^* = 1 - \frac{2k}{q}, \quad y_R^* = \frac{2k}{q}, \quad (12)$$

*and exists for  $0 \leq k \leq q/2$ . It resists an infinitesimal ALLD invasion if*

$$k > \frac{cq}{b+c}. \quad (13)$$

3. *Whenever the two boundary equilibria are the relevant stable outcomes, their ALLC frequencies are equal at*

$$k^*(\mu_0) = \sqrt{\frac{q\mu_0}{2}}. \quad (14)$$

*For  $cq/(b+c) < k < k^*(\mu_0)$ , the responsive equilibrium contains more ALLC than the constant-switching equilibrium.*

*Proof.* On  $z = 0$ , let  $y = 1 - x$ . For constant switching, the ALLC equation reduces to

$$\dot{x} = x \left( \frac{q}{2} y^2 - \mu_0 \right), \quad (15)$$

which gives (10). At that equilibrium the mean payoff is  $q - \mu_0$ , so the growth rate of rare ALLD is  $b(1 - y_F^*) - (q - \mu_0)$ , yielding (11).

For responsive switching,  $\mu_0 z = 0$  on the boundary. Coexistence requires equality of the cost-adjusted ALLC and TFT payoffs:  $q - k = q(1 - y/2)$ . This gives (12). The growth rate of rare ALLD is

$$b \left( 1 - \frac{2k}{q} \right) - (q - k) = c - \frac{b+c}{q}k, \quad (16)$$

which is negative precisely under (13). Finally, equating (10) and (12) gives (14); the strict comparison follows from the same expressions.  $\square$

For  $b = 5$  and  $c = 1$ , (14) becomes  $k^* = \sqrt{2\mu_0}$ , while transverse stability of the responsive boundary requires  $k > 2/3$ . At the break-even value this condition holds when  $\mu_0 > 2/9$ . The constant boundary is transversely stable when  $g_F(\mu_0) < 0$  and exists through  $\mu_0 = 2$ .

## 4 Numerical Methods

The deterministic system was integrated from  $(x, y, z) = (1/3, 1/3, 1/3)$  to  $t = 800$  with an eighth-order adaptive solver, relative tolerance  $2 \times 10^{-11}$ , and absolute tolerance  $2 \times 10^{-13}$ . The implementation neither clips negative values nor renormalizes the state after integration; terminal states were checked against the simplex constraint.

The continuous-time jump process was simulated exactly by sampling the next event from the total birth and switching rates. Each trial began with  $N - 1$  ALLC individuals and one ALLD individual. Rescue occurred when ALLD first reached extinction, failure occurred when ALLD fixed, and pre-rescue observation was capped after  $1000N$  birth-death events. No reported trial was censored. After rescue, composition was recorded after  $10N, 20N, \dots, 100N$  additional birth-death events; switch events occurring between those events remained part of the continuous-time process.

The main sweep used  $N = 100$ ,  $b = 5$ ,  $c = 1$ ,  $w = 1$ , 11 values of  $\mu_0$  from 0 to 2, and 500 independent trials per point. Robustness was evaluated at  $\mu_0 = 1$  for all 27 combinations of  $N \in \{50, 100, 200\}$ ,  $w \in \{0.25, 0.5, 1\}$ , and  $b/c \in \{3, 5, 7\}$  with  $c = 1$  and 300 trials per combination. The cost sweep used  $N = 100$ ,  $b = 5$ ,  $c = 1$ ,  $w = 1$ ,  $\mu_0 = 1$ ,  $k \in \{0, 0.2, \dots, 2\}$ , and 400 trials per nonzero cost. Means are shown with normal-approximation 95% confidence intervals. Source code, fixed random seeds, and numerical output are provided in the repository.

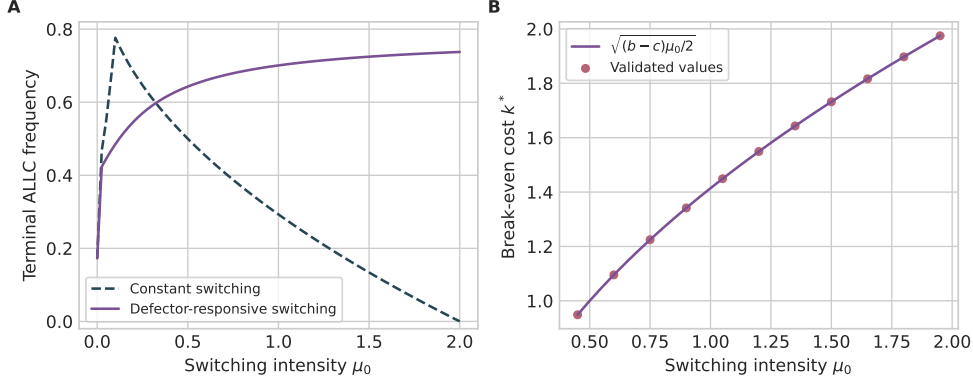
## 5 Results

### 5.1 Deterministic comparison and cost threshold

Figure 1A shows that constant switching progressively removes ALLC as  $\mu_0$  increases, while responsive switching rises to a high-ALLC equilibrium over the plotted range. At  $\mu_0 = 0.5$ , the terminal ALLC frequencies are 0.500 under constant switching and 0.643 under responsive switching; at  $\mu_0 = 1$ , they are 0.293 and 0.701, respectively.

At  $\mu_0 = 2$ , the constant-switching boundary reaches zero ALLC, while the responsive trajectory has terminal ALLC frequency 0.738.

For the costly model, we evaluated (14) at 31 values of  $\mu_0$  from 0.45 to 1.95. Figure 1B displays the predicted curve. The maximum absolute difference between the numerically integrated costly-responsive ALLC equilibrium and its constant-switching counterpart was  $2.28 \times 10^{-11}$ . This check supports the implementation, while the threshold itself follows analytically from the proposition.

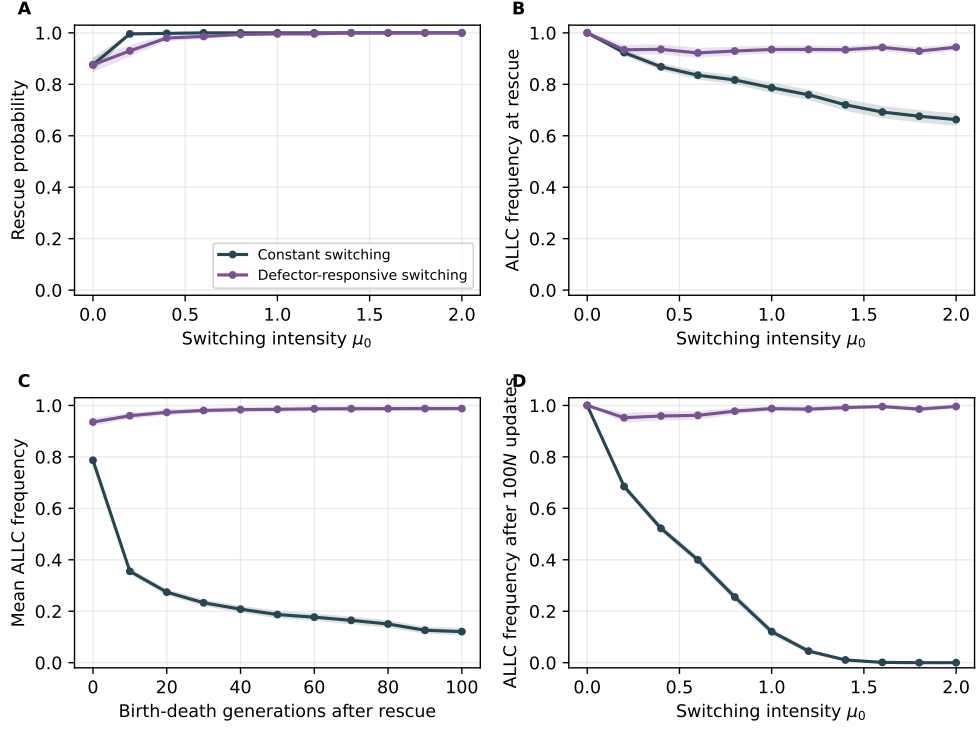


**Fig. 1** Deterministic comparison for  $b = 5$  and  $c = 1$ . (A) Terminal ALLC frequency from an interior initial condition under constant and defector-responsive switching. (B) Exact break-even readiness cost from (14); markers show parameters checked by numerical integration.

## 5.2 Finite-population tradeoff and post-rescue horizon

Figure 2 separates rescue probability, composition at rescue, and composition over time after rescue. At low intensity, constant switching rescues more reliably because it creates TFT without waiting for a strong defector signal. At  $\mu_0 = 0.2$ , rescue probabilities are 0.996 for constant switching and  $0.930 \pm 0.022$  for responsive switching. By  $\mu_0 = 1$ , both are near one: 1.000 and  $0.996 \pm 0.006$ .

The post-rescue comparison reverses. At  $\mu_0 = 1$ , the mean ALLC frequencies at rescue are  $0.787 \pm 0.008$  and  $0.935 \pm 0.005$  for constant and responsive switching. After  $10N$  additional birth-death events they are  $0.355 \pm 0.013$  and  $0.960 \pm 0.009$ ; after  $50N$ ,  $0.187 \pm 0.015$  and  $0.985 \pm 0.007$ ; and after  $100N$ ,  $0.121 \pm 0.014$  and  $0.988 \pm 0.007$ . Constant switching continues to convert ALLC to TFT after  $z = 0$ . Responsive switching stops immediately, after which selection and drift on the ALLC-TFT edge generally restore ALLC. The time course therefore reflects a structural difference, not a transient created by selecting a single horizon.



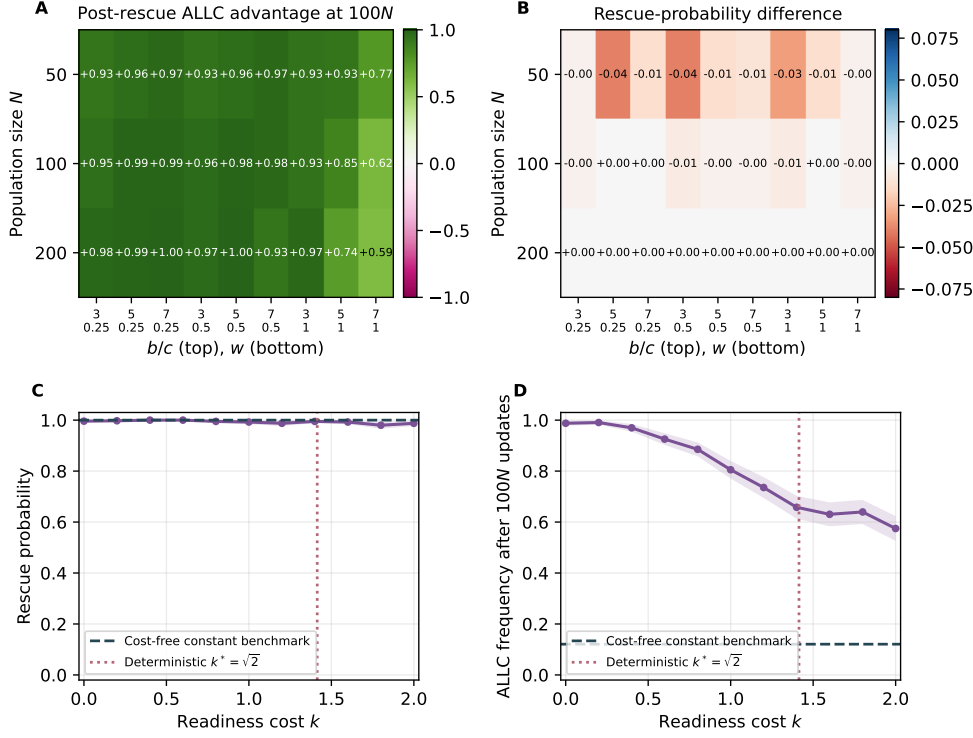
**Fig. 2** Continuous-time finite-population results for  $N = 100$ ,  $b = 5$ ,  $c = 1$ ,  $w = 1$ , and 500 trials per point. Lines show means and shading shows 95% confidence intervals. (A) Probability that ALLD reaches extinction before fixation. (B) ALLC frequency at first ALLD extinction. (C) Post-rescue time course at  $\mu_0 = 1$ . One demographic generation is  $N$  birth-death events. (D) ALLC frequency after  $100N$  post-rescue birth-death events.

### 5.3 Robustness and readiness cost

The preservation advantage survives all requested parameter changes (Figure 3A). Across the 27 combinations of  $N$ ,  $w$ , and  $b/c$ , the responsive-minus-constant difference in mean ALLC frequency after  $100N$  post-rescue birth-death events ranges from 0.594 to 0.999. This advantage is not produced by a large rescue-probability benefit. Responsive switching has a rescue probability between 0 and 0.040 below the constant rule across the same combinations (Figure 3B). Thus, the robust tradeoff is slightly weaker or equal rescue paired with substantially better post-rescue preservation.

Including  $k$  in the finite process connects the stochastic comparison to the analytical result. At  $\mu_0 = 1$ , the deterministic break-even value is  $k^* = \sqrt{2} \approx 1.414$ . As  $k$  rises from 0 to 2, responsive rescue probability remains between 0.980 and 1.000, but its mean ALLC frequency after  $100N$  events falls from 0.988 at  $k = 0$  to 0.658 at  $k = 1.4$  and 0.575 at  $k = 2$  (Figure 3C,D). The cost-free constant benchmark is 0.121 at the same horizon. The deterministic threshold is not expected to be an exact crossing point for this conditional finite-time statistic, but it correctly identifies the parameter scale on which readiness cost materially erodes the responsive advantage.





**Fig. 3** Robustness and readiness cost. (A,B) Responsive-minus-constant differences at  $\mu_0 = 1$  across 27 combinations of population size, selection strength, and benefit-to-cost ratio; values are based on 300 trials per condition. (A) ALLC frequency after  $100N$  post-rescue birth-death events. (B) Rescue probability. (C,D) Cost sweep for the responsive process with 400 trials per nonzero cost. Dashed lines show the cost-free constant benchmark and dotted lines show the deterministic threshold  $k^* = \sqrt{2}$ . (C) Rescue probability. (D) ALLC frequency after  $100N$  events.

## 6 Discussion

The defector-proportional response is an appealing negative-feedback mechanism, but it is not new to this paper: [Pal et al. \(2025\)](#) presented it in Supplementary Figure S8. The present study adds a general cost theorem with stability restrictions, a continuous-time individual process with a derived deterministic limit, and a robustness analysis across demographic and game parameters. These additions convert the finite-population comparison from a reproduction-linked mutation rule into a model of individual phenotype change and clarify which conclusions depend on time horizon.

The distinction from earlier finite-population work is important. [Imhof et al. \(2007\)](#) studied mutation-selection dynamics among ALLC, ALLD, and TFT and emphasized the vulnerability of TFT to drift toward ALLC. Here, transition is one-way and activated by current ALLD abundance; the question is not which strategy dominates under recurrent mutation, but how a temporary responsive phenotype can rescue and then preserve an ALLC population. Likewise, game-environment feedback changes payoffs as an environmental state evolves ([Weitz et al. 2016](#)), whereas the present feedback

changes only the ALLC-to-TFT transition intensity. The information assumption is closer to perceptual evolutionary models in which decisions depend on social or environmental signals (Salahshour 2025), although our signal is the exact global frequency  $z$ .

The post-rescue horizon cannot be omitted. Once ALLD disappears, responsive switching stops but constant one-way switching continues. In a finite population, the long-run absorbing structures therefore differ: responsive dynamics on the ALLC-TFT edge ultimately reaches one of its monomorphic endpoints, while constant switching makes all-ALLC nonabsorbing and favors eventual loss of ALLC. Results at  $10N$ ,  $50N$ , and  $100N$  show how this structural difference emerges rather than presenting a single selected endpoint.

The readiness cost is intentionally specific. It represents the opportunity or maintenance cost of remaining in an ALLC state that is prepared to detect defectors and activate TFT. If a biological system instead pays a persistent sensing cost after activation, both the payoff model and the boundary theorem must change. The stochastic cost sweep shows that the state-specific cost can strongly reduce post-rescue ALLC even when rescue remains likely, so cost cannot be ignored when evaluating the mechanism.

Several limitations remain. The response uses instantaneous global knowledge of  $z$ ; local, delayed, or noisy perception could change activation. The payoff matrix is a reduced limit of an infinitely repeated donation game with symmetric rare implementation errors, not the standard noiseless repeated Prisoner’s Dilemma. The common birth-rate baseline  $a$  affects demographic event speed and finite-population variance even though it cancels from deterministic drift. Finally, the model contains only one switching direction. A biologically specific application would require a mechanistic sensing timescale, local information, a persistent-cost decision, and parameters tied to data.

## 7 Conclusion

Defector-responsive switching trades slightly weaker rescue at low signal strength for substantially better preservation of unconditional cooperation after rescue. The exact deterministic break-even cost is  $k^*(\mu_0) = \sqrt{(b-c)\mu_0}/2$  subject to explicit existence and invasion conditions. A continuous-time Moran-type process with separate individual transitions converges directly to the deterministic equations and gives the switching intensity an unambiguous timescale and phenotype interpretation. The resulting finite-population tradeoff persists across population size, selection strength, benefit-to-cost ratio, post-rescue horizon, and readiness cost.

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### Competing Interests

The author has no relevant financial or non-financial interests to disclose.

### Author Contributions

Aarush Gunjal performed the conceptualization, methodology, formal analysis, software development, numerical investigation, visualization, and manuscript preparation. The author read and approved the final manuscript.

### Data Availability

The numerical data supporting the findings of this study are available in the public repository at [https://github.com/aarushgunjal/defector\\_responsive\\_strategy\\_switching](https://github.com/aarushgunjal/defector_responsive_strategy_switching).

### Code Availability

The source code and figure-generation materials needed to reproduce the reported results are available at [https://github.com/aarushgunjal/defector\\_responsive\\_strategy\\_switching](https://github.com/aarushgunjal/defector_responsive_strategy_switching).

### Use of Generative AI

Generative AI tools were used for language and presentation assistance during manuscript preparation. The author independently reviewed and verified the mathematical analysis, code, citations, numerical results, and final text and accepts full responsibility for the manuscript.

## References

- Axelrod, R., Hamilton, W.D.: The evolution of cooperation. *Science* **211**, 1390–1396 (1981) <https://doi.org/10.1126/science.7466396>
- Antal, T., Traulsen, A., Ohtsuki, H., Tarnita, C.E., Nowak, M.A.: Mutation-selection equilibrium in games with multiple strategies. *Journal of Theoretical Biology* **258**, 614–622 (2009) <https://doi.org/10.1016/j.jtbi.2009.02.010>
- Axelrod, R.: *The Evolution of Cooperation*. Basic Books, New York (1984)
- Bergin, J., Lipman, B.L.: Evolution with state-dependent mutations. *Econometrica* **64**, 943–956 (1996) <https://doi.org/10.2307/2171851>

- Chen, L., Deng, C., Duong, M.H., Han, T.A.: On the number of equilibria of the replicator-mutator dynamics for noisy social dilemmas. *Chaos, Solitons & Fractals* **180**, 114565 (2024) <https://doi.org/10.1016/j.chaos.2024.114565>
- Fudenberg, D., Imhof, L.A.: Imitation processes with small mutations. *Journal of Economic Theory* **131**, 251–262 (2006) <https://doi.org/10.1016/j.jet.2005.04.006>
- Hadeler, K.P.: Stable polymorphisms in a selection mutation model. *SIAM Journal on Applied Mathematics* **41**, 1–7 (1981) <https://doi.org/10.1137/0141001>
- Hofbauer, J.: The selection mutation equation. *Journal of Mathematical Biology* **23**, 41–53 (1985) <https://doi.org/10.1007/BF00276557>
- Hofbauer, J., Sigmund, K.: *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge (1998). <https://doi.org/10.1017/CBO9781139173179>
- Imhof, L.A., Fudenberg, D., Nowak, M.A.: Evolutionary cycles of cooperation and defection. *Proceedings of the National Academy of Sciences* **102**, 10797–10800 (2005) <https://doi.org/10.1073/pnas.0502589102>
- Imhof, L.A., Fudenberg, D., Nowak, M.A.: Tit-for-tat or win-stay, lose-shift? *Journal of Theoretical Biology* **247**(3), 574–580 (2007) <https://doi.org/10.1016/j.jtbi.2007.03.027>
- Kurtz, T.G.: Solutions of ordinary differential equations as limits of pure jump markov processes. *Journal of Applied Probability* **7**(1), 49–58 (1970) <https://doi.org/10.2307/3212147>
- Moran, P.A.P.: *The Statistical Processes of Evolutionary Theory*. Clarendon Press, Oxford (1962)
- Nowak, M.A.: Five rules for the evolution of cooperation. *Science* **314**, 1560–1563 (2006) <https://doi.org/10.1126/science.1133755>
- Nowak, M.A., Sigmund, K.: Tit for tat in heterogeneous populations. *Nature* **355**, 250–253 (1992) <https://doi.org/10.1038/355250a0>
- Nowak, M.A., Sasaki, A., Taylor, C., Fudenberg, D.: Emergence of cooperation and evolutionary stability in finite populations. *Nature* **428**, 646–650 (2004) <https://doi.org/10.1038/nature02414>
- Pal, S., Lambert, A., Nowak, M.A.: Stabilizing unconditional cooperation. *Proceedings of the Royal Society A* **481**, 20240945 (2025) <https://doi.org/10.1098/rspa.2024.0945>
- Page, K.M., Nowak, M.A.: Unifying evolutionary dynamics. *Journal of Theoretical Biology* **219**, 93–98 (2002) <https://doi.org/10.1006/jtbi.2002.3066>
- Salahshour, M.: Perceptual rationality: an evolutionary game theory of perceptually rational decision-making. *Royal Society Open Science* **12**(10), 251125 (2025) <https://doi.org/10.1098/rsos.251125>

- Stewart, A.J., Plotkin, J.B.: From extortion to generosity, evolution in the iterated prisoner's dilemma. *Proceedings of the National Academy of Sciences* **110**, 15348–15353 (2013) <https://doi.org/10.1073/pnas.1306246110>
- Tkadlec, J., Hilbe, C., Nowak, M.A.: Mutation enhances cooperation in direct reciprocity. *Proceedings of the National Academy of Sciences* **120**, 2221080120 (2023) <https://doi.org/10.1073/pnas.2221080120>
- Taylor, P.D., Jonker, L.B.: Evolutionarily stable strategies and game dynamics. *Mathematical Biosciences* **40**, 145–156 (1978) [https://doi.org/10.1016/0025-5564\(78\)90077-9](https://doi.org/10.1016/0025-5564(78)90077-9)
- Traulsen, A., Nowak, M.A., Pacheco, J.M.: Stochastic dynamics of invasion and fixation. *Physical Review E* **74**, 011909 (2006) <https://doi.org/10.1103/PhysRevE.74.011909>
- Weitz, J.S., Eksin, C., Paarporn, K., Brown, S.P., Ratcliff, W.C.: An oscillating tragedy of the commons in replicator dynamics with game-environment feedback. *Proceedings of the National Academy of Sciences* **113**(47), 7518–7525 (2016) <https://doi.org/10.1073/pnas.1604096113>